D - Minimize Range

分值：400 分

Problem Statement

You are given a sequence A of N positive integers and a positive integer K.
You can perform the following operation on the sequence A any number of times.

给定一个由 N 个正整数组成的序列 A，以及一个正整数 K。 你可以对序列 A 执行任意次以下操作。

- Choose an integer i with $1 \leq i \leq N$, and add K to A_i.
- 选择一个满足 $1 \leq i \leq N$ 的整数 i，将 K 加到 A_i 上。

Find the minimum possible value of $\max(A) - \min(A)$.

求 $\max(A) - \min(A)$ 的最小可能值。

Constraints

- $1 \leq N \leq 2 \times 10^5$
- $1 \leq K \leq 10^9$
- $1 \leq A_i \leq 10^9$
- All input values are integers.
- $1 \leq N \leq 2 \times 10^5$
- $1 \leq K \leq 10^9$
- $1 \leq A_i \leq 10^9$
- 所有输入值均为整数。

Input

The input is given from Standard Input in the following format:

输入以如下格式从标准输入给出：

N K
A_1 A_2 $\ldots$ A_N

Output

Output the answer on a single line.

输出答案，占一行。

Sample Input 1

```
3 10
3 21 9
```

Sample Output 1

```
4
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First, choosing $i = 1$ makes the sequence $A = (13, 21, 9)$.
Next, choosing $i = 3$ makes the sequence $A = (13, 21, 19)$.
Next, choosing $i = 1$ makes the sequence $A = (23, 21, 19)$.
At this point, $\max(A) - \min(A) = 23 - 19 = 4$.
It is impossible to make $\max(A) - \min(A)$ at most 3, so the answer is 4.

首先，选择 $i = 1$，使得序列变为 $A = (13, 21, 9)$。 接下来，选择 $i = 3$，使得序列变为 $A = (13, 21, 19)$。 然后，选择 $i = 1$，使得序列变为 $A = (23, 21, 19)$。 此时，$\max(A) - \min(A) = 23 - 19 = 4$。 无法使 $\max(A) - \min(A)$ 不超过 3，因此答案是 4。

Sample Input 2

```
5 6
4 100 5 10 450
```

Sample Output 2

2